

Akshay Mahale

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Profile Summary

- 3.8 years of hands-on experience in analytics, problem-solving, and delivering actionable solutions across diverse business challenges
- Proficient in **Generative AI workflows**, including **data ingestion**, **text processing**, **chunking**, and **semantic embedding** using tools like **PyMuPDF**, **Tesseract**, and **Sentence-BERT**. Experienced in leveraging vector databases like **Pinecone** and **Qdrant** for efficient information retrieval and **fine-tuning models** such as **Llama 2** and **Gemini** for context-aware, natural language responses.
- Expertise in advanced ML techniques such as **Regression**, **SVM**, **Random Forest**, **Ensembles**, **Clustering**, and **Gradient Boosting**, with proficiency in **Python** and **TensorFlow** for model development and deployment.
- Strong communicator** with the ability to bridge technical expertise and business needs, ensuring **stakeholder alignment** and timely delivery of **impactful AI solutions**.

Core Competencies

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|--------------------|------------------|---------------|-------------------------|
| Machine Learning | Deep Learning | Generative AI | Artificial Intelligence |
| Evaluation Metrics | Vector Databases | LLM | Time Series Analysis |
| Data Analytics | Cloud Computing | NLP | Agentic AI |

Experience

Mobicool Software Solutions Pvt Ltd – Associate Data Scientist Oct 2022 – Apr 2026

- Developed and implemented **AI solutions leveraging the RAG (Retrieval-Augmented Generation) framework**, transforming large-scale unstructured enterprise data into actionable insights for internal and client-facing applications.
- Built and optimized **NLP pipelines** involving text cleaning, tokenization, embedding generation, and semantic retrieval to support intelligent knowledge systems and document understanding workflows.
- Managed **end-to-end data pipelines**, including data ingestion, preprocessing, workflow automation, vector storage, and semantic search integration using tools such as **Microsoft Graph API**, **n8n**, and **Qdrant**, ensuring accuracy, scalability, and data integrity.
- Collaborated with cross-functional teams to **enhance AI-driven automation**, improve information accessibility, streamline workflows, and support data-driven decision-making across multiple departments.

Project

AI-Contextual Medical Query Framework | *RAG, Transformers, NLP, Embeddings, Evaluation* Domain - Healthcare

- Implemented an end-to-end **RAG (Retrieval-Augmented Generation)** framework to build an intelligent **query-response** system, combining retrieval and generation capabilities.
- Developed and optimized pipelines for **data ingestion**, **chunking**, **semantic embeddings (using Sentence-BERT)**, and **vector storage** in **Pinecone** to enable efficient and accurate semantic search.
- Integrated Llama 2 for contextual answer generation**, fine-tuned for domain-specific responses, and enhanced system performance using **Evaluation metrics** like **BLEU**, **ROUGE**, **precision**, and **recall**.

AI-Financial Aid Knowledge Assistant System | *RAG, Microsoft Graph API, Qdrant, n8n* Domain - Education

- Developed an end-to-end **RAG (Retrieval-Augmented Generation)** pipeline to transform enterprise SharePoint content into an intelligent, query-responsive knowledge assistant for internal staff.
- Built automated pipelines for **document crawling**, **cleaning (ASPX- Markdown)**, **chunking**, **semantic embeddings**, and vector storage in **Qdrant** to enable accurate semantic search across large, heterogeneous document sets.
- Integrated workflows using **n8n**, **Microsoft Graph API**, and **Azure AD** authentication to enable scheduled ingestion, monitoring, and automated updates, improving assistant reliability and knowledge freshness.

AI-Powered Customer Retention Platform | *Numpy, Pandas, Seaborn, Matplotlib, Python* Domain - Telecommunication

- Developed predictive models with **85% accuracy** to identify high-risk customers, achieving a **20% reduction in churn** over 12 months. Implemented cost-effective retention strategies, lowering the **annual churn rate by 15%** and enhancing customer loyalty. Optimized customer acquisition costs by prioritizing satisfaction and long-term retention.
- Managed data collection, preprocessing, **feature engineering**, and **imbalanced dataset handling (using SMOTE)** for robust **predictive analysis**, enabling actionable insights from structured data.

- Designed and fine-tuned **Random Forest** models using **hyperparameter tuning** and **regularization techniques** to achieve optimal performance. Deployed the solution on **AWS EC2**, ensuring scalability and real-time predictions.

Technical Skills

Analytical Tools: Advanced Excel, MySQL, Spyder, Jupyter, Google collab, Pyspark, Git, Jira, VS code, n8n, Postman, Ryze

Languages : Python, SQL

Frameworks/ Libraries: Tensorflow, Pytorch, Pandas, Numpy, Sci-kit learn, nltk, matplotlib, Keras, Spacy, PyMuPDF, Pdfplumber, Tesseract, Sentence-BERT, Llama 2, Pinecone, Qdrant, Flask, Fast API, MongoDB, LangChain, LangGraph, Hugging Face

Technical Proficiencies: EDA, PCA, Hyperparameter tuning, Regularization, Cross Validation, Evaluation Metrics, Standardization, SMOTE, Confusion Matrix, VIF, Streamlit, Transformers (Llama 2, GPT), Word embedding, TF-IDF, BOW, Text mining, Tokenization, Lemmatization, Stemming, LSTM, RNN, ANN, CNN, POS tagging, TextBlob, AWS Cloud, Neural Network, Encoder-Decoder Transformer Architecture, Attention Mechanism, AI Agents, Agentic AI, Azure AD, Microsoft Graph API, Rest API, Web Crawling.

Education

Savitribai Phule Pune University

Bachelor of Engineering – CGPA: 7.46

June 2021

Nasik, Maharashtra

Extracurricular / Certifications

- Full Stack Data Science, Udemy.
- Google Data Analytics, Coursera.
- Supply chain planning and scheduling at TCS, Coursera.
- Applied generative AI, Simplilearn.